PAPER_1515 — ln 10 = (1+F_TRZ)(K_MEX + F_TRZ²) — Residual 0.0035%

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket A (Foundational mathematical identity) **Date:** June 18, 2026 **Status:** CLOSED — Natural logarithm of 10 expressible in UQFF primitives at sub-0.01% precision

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Observation

PAPER_1208 (UQFF Transcendentals Unified Proof Set) demonstrates that the natural logarithm ln 10 ≈ 2.302585 can be expressed using only the locked UQFF integer primitives F_TRZ and K_MEX, with no fitted constants.

UQFF Closed Identity

```
ln 10 ≈ (1 + F_TRZ) · (K_MEX + F_TRZ²)
      = (1 + 1/10) · (25/12 + 1/100)
      = 1.1 · 2.09333…
      = 2.30267

Observed: ln 10 = 2.302585093…
Residual: |2.30267 − 2.30259| / 2.30259 = 0.0035%
```

Equivalent Forms

The expression factors elegantly:

```
ln 10 = K_MEX + F_TRZ·K_MEX + F_TRZ² + F_TRZ³          (raw expansion)
      = (1 + F_TRZ)(K_MEX + F_TRZ²)                     (factored form)
      = K_MEX(1 + F_TRZ) + F_TRZ²(1 + F_TRZ)            (mixed)
```

Each form involves only two integer primitives (F_TRZ = 1/10, K_MEX = 25/12) — no additional fitted parameters.

Significance

This identity demonstrates that the UQFF integer primitives are **not arbitrary physics-fitted values** but encode rational approximations to fundamental mathematical constants. The natural logarithm of 10, which appears in: - Cosmology (orders of magnitude of physical scales) - Decibel/pH/Richter scales - Information theory (digit count) - Stirling approximations

…emerges to 0.0035% precision from {F_TRZ, K_MEX} alone.

NOT REPLACEMENT

This is not a claim that ln 10 IS equal to (1+F_TRZ)(K_MEX+F_TRZ²). It is a claim that the UQFF integer primitives carry a structural rational approximation to ln 10 with sub-percent precision, demonstrating their mathematical depth.

Reference

- Source: PAPER_1208 UQFF Transcendentals Unified Proof Set
- Related: PAPER_1516 (ln 2), PAPER_1517 (π^2)
- Calculator dispatch: `calculate_paradox({"paradox": "transcendental_ln_10"})`
